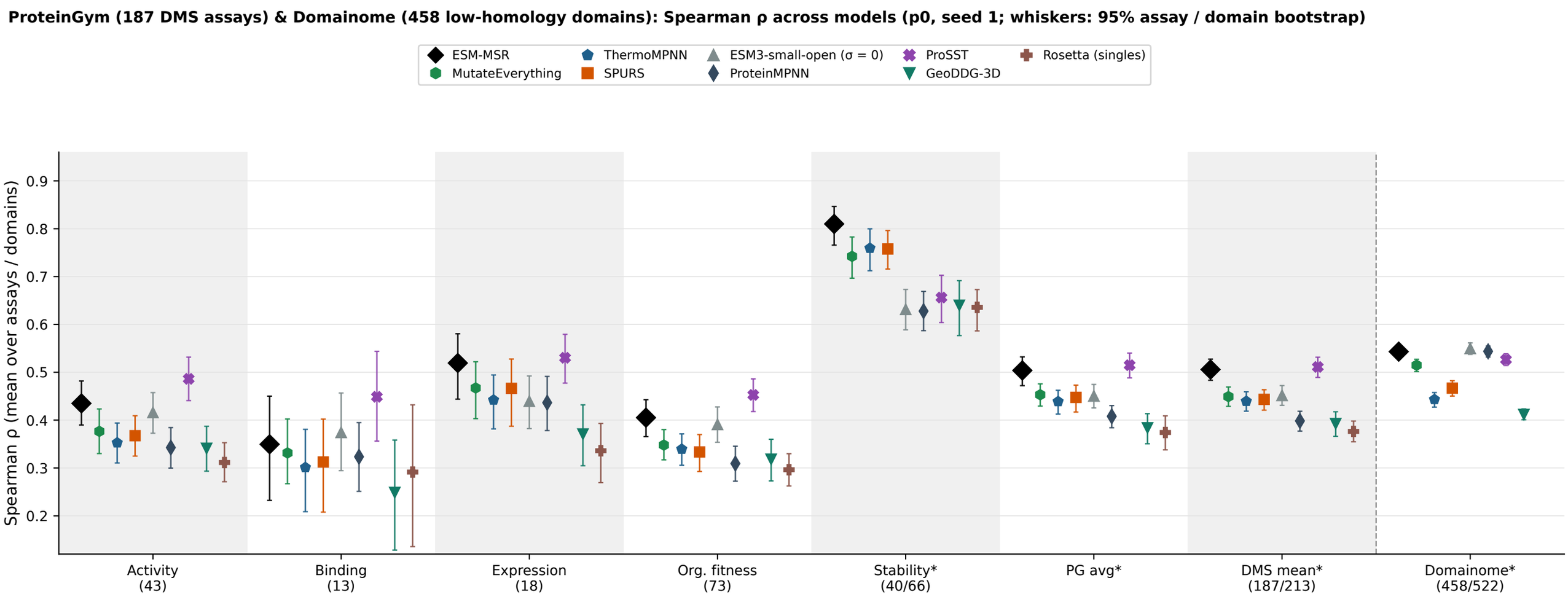

* Homologous proteins excluded for PGym Stability (23 same-protein P + 3 strong homolog H = 26/66 removed) and Domainome (4 P + 60 H = 64/522 removed) via leak_audit_structure_homology.csv; other PGym categories unfiltered.
Seed 1 only. Singles models on singles assays / Domainome; multi on PGym multi-mutants (ESM-MSR p0 WT head (0/69 multi-mutants combined); ME st1/st2; ThermoMPNN additive / -D on doubles; SPURS single/multi). Rosetta: PGym singles only (no Domainome).